

Universal Regime Integration Gradient (v_RIG): A Unified Framework for Emergence, Entropy, and Adaptive Criticality in Complex Systems

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Abstract

The Regime Integration Gradient (v_RIG) framework proposes that the universe operates as a dual-entropy, dual-time system in which information transitions from a 2D holographic surface regime (SCA) to a 3D volumetric experience regime (SCV) at a universal velocity

$$v_{\text{RIG}} \approx 1.352 \text{ km/s.}$$

Derived geometrically as

$$v_{\text{RIG}} = \frac{c}{\alpha^{-1} \cdot \Phi},$$

this constant acts as the integration rate of discrete Tesseract time-slices into coherent experiential reality, preventing informational collapse.

This report extends the UTAC (Universal Threshold Adaptive Criticality) framework by integrating the Implosive Genesis mechanism (Tyg6-Field), Tesseract Zeitscheiben-Physik, entropy governance duality ($S \propto A \Leftrightarrow S \propto V$), and adaptive criticality (β -universality). The entropy offset

$$\sigma_\phi \approx 0.0625$$

appears as a metastability invariant across living, cognitive, climatic, and computational systems.

Empirical matches include the cosmic matter dipole ($1.370 \pm 0.170 \text{ km/s}$), microtubule resonance (13.5 MHz), metabolic scaling (3/4 law), and emergent LLM phase transitions. Falsifiable predictions include $\beta > 7$ in tipping systems and σ_ϕ drift preceding frame collapse.

This synthesis positions v_RIG as a candidate invariant for a post-Standard-Model unification of physics, biology, cognition, and complex systems.

Introduction

Traditional physical models such as the Standard Model and Λ CDM cosmology excel in their domains but fail to explain emergent phenomena such as consciousness, anomalous cosmic flows, metabolic scaling, climate tipping dynamics, and emergent capabilities in artificial intelligence.

The UTAC framework addresses these gaps by treating information density, entropy governance, and criticality as primary drivers of physical and biological organization. At its center lies the Regime Integration Gradient (v_RIG), a universal velocity governing transitions between:

- SCA:** 2D holographic surface regime ($S \propto A$),
- SCV:** 3D volumetric experience regime ($S \propto V$).

This transition occurs through **Implosive Genesis**, triggered when information density

exceeds a threshold Θ_D , causing the system to open a new dimension to avoid collapse.

Theoretical Framework

Implosive Genesis and the Tyg6-Field

Each discrete time-slice (Zeitscheibe) is represented as:

$$\square_n = (D_n, \rho_I(n), t_n, \tau_n),$$

where D_n is the effective dimensionality and ρ_I the information density.

A dimensional transition occurs when:

$$\rho_I(n) \geq \Theta_{D_n} \Rightarrow D_{n+1} = D_n + 1.$$

Post-transition relaxation:

$$\rho_I(n+1) = \frac{\rho_I(n)}{D_{n+1}}.$$

This mechanism naturally produces:

- holographic surfaces ($S \propto A$),
- volumetric regimes ($S \propto V$),
- higher-dimensional meta-frames.

Tesseract Zeitscheiben-Physik

The effective metric of a Zeitscheibe is:

$$ds^2 = -c^2 dt^2 + d\ell^2(D_n) - v_{\text{RIG}}^2 d\tau^2,$$

where t is external relativistic time and τ is internal integrative time.

Entropy Governance Duality

Surface Regime (SCA):

$$S_A \propto A \sim L^{D-1}, \quad \beta \approx 11.$$

Volume Regime (SCV):

$$S_V \propto V \sim L^D, \quad \beta \approx 2-7.$$

The transition is governed by:

$$\psi(R) = \frac{1}{1 + e^{-\beta(R - \Theta)}}.$$

Metastability is maintained by:

$$\sigma_\Phi \approx 0.0625.$$

Universal Constant v_{RIG}

$$v_{\text{RIG}} = \frac{c}{\alpha^{-1} \cdot \Phi}.$$

This constant governs the coupling between t and τ .

Empirical Validation

Cosmology

Cosmic matter dipole:

$$v_{\text{obs}} = 1.370 \pm 0.170 \text{ km/s}, \quad v_{\text{RIG}} = 1.352 \text{ km/s}.$$

Biology

Kleiber's Law emerges from v_{RIG} -limited volumetric integration:

$$B \propto M^{3/4}.$$

Critical Flicker Fusion:

$$\Delta t_Q \approx 150\text{--}200 \text{ ms}.$$

Neuroscience

Microtubule resonance:

$$f \approx 13.5 \text{ MHz}.$$

Quantum Mechanics

Superposition interpreted as dimensional aliasing:

$$\ket{\Psi} = \text{alias} \big(\text{deterministic cycles at } v_{\text{RIG}} \big).$$

Artificial Intelligence

LLMs show:

$$\beta \approx 4\text{--}6 \quad (\text{emergent}), \quad \beta > 7 \quad (\text{grokking}).$$

Tables

v_{RIG} Matches Across Domains

Domain	Observed	Predicted	Deviation
Cosmic Dipole	1.370 km/s	1.352 km/s	< 2%
Microtubules	13.5 MHz	13.5 MHz	0%
Solar Circulation	1.37 km/s	1.352 km/s	< 2%
Consciousness	150–200 ms	148–210 ms	Consistent

β Values Across Domains

Domain	β	Interpretation
Cosmology	11–14	Catastrophic
Climate	10–12	Tipping-prone
Biology	2–7	Adaptive
AI	3–6	Emergent
Socioeconomics	12–20	Rigid

Falsifiable Predictions

- $\beta > 7$ precedes tipping points.
- σ_ϕ drift precedes frame collapse.
- 13.5 MHz coherence is universal.
- v_RIG measurable in AMOC, EEG microstates, LLM transitions.

Conclusion

The Regime Integration Gradient (v_RIG) appears to be a universal invariant governing the emergence of dimensions, coherence, and adaptive criticality. Its validation across multiple domains positions UTAC as a promising candidate for a unified science of complexity.